

Product Line Profitability & Margin Performance Analysis for Nassau Candy Distributor

1. Introduction

In today's competitive business environment, sales volume alone is not sufficient to measure business success. A product may generate high sales but contribute relatively little profit because of high manufacturing costs or low margins. Conversely, a product with lower sales volume may generate a strong gross margin and significant profit per unit. Therefore, organizations need a detailed profitability analysis to understand which products and divisions are genuinely contributing to financial performance.

This project, "Nassau Candy Product Line Profitability Analysis," focuses on evaluating the profitability of Nassau Candy's product portfolio using transactional sales, cost, unit, product, division, and geographic data. The analysis examines the relationship between sales, manufacturing cost, gross profit, gross margin, units sold, and profit contribution to provide a comprehensive view of product-level and division-level financial performance.

The project specifically addresses the business need to identify high-profit and high-margin products, high-sales but low-margin products, underperforming divisions, cost-heavy products, and products that may require pricing or cost optimization. In addition, Pareto analysis is used to determine whether the company's revenue and profit are concentrated among a small number of products, helping identify potential portfolio dependency risks.

A structured analytical methodology was implemented using Python, Pandas, NumPy and Plotly, followed by the development of an interactive Streamlit dashboard. The dashboard enables users to explore profitability results through date-range selection, division filtering, margin thresholds, and product search.

The analysis ultimately transforms raw transactional data into actionable business insights that can support pricing decisions, cost renegotiation, product portfolio management, and profitability improvement.

2. Background

Nassau Candy operates in a business environment where product sales volume alone may not accurately represent financial performance. A product can generate substantial revenue while contributing relatively little profit because of high manufacturing costs, pricing pressure or low margins.

Similarly, products with comparatively low sales may have strong margins and attractive profit-per-unit economics. Therefore, management needs visibility not only into what sells, but also into **what generates sustainable profit**.

The profitability analysis focuses on understanding the relationship between:

- Sales revenue
- Manufacturing cost
- Gross profit
- Gross margin
- Units sold
- Profit per unit

- Product contribution
- Division performance
- Geographic concentration

The objective is to move beyond simple sales reporting and develop a **profitability-driven view of the product portfolio**.

3. Problem Statement

The organization currently lacks sufficient visibility into the financial performance of individual products and product divisions.

The key business questions addressed by this project are:

Product-level questions

- Which products generate the highest gross profit?
- Which products have the highest gross margins?
- Are high-sales products also financially efficient?
- Which products generate high sales but relatively low margins?
- Which products generate low sales and low profit?
- Which products may require repricing or cost optimization?

Division-level questions

- Which product divisions generate the greatest revenue?
- Which divisions generate the greatest gross profit?
- How does gross margin vary across divisions?
- Are there divisions where revenue is relatively strong but profitability is weak?
- Which divisions demonstrate stronger financial efficiency?

Cost-related questions

- How strongly are sales associated with manufacturing cost?
- Which products have high cost relative to sales?
- Which products have cost and margin characteristics that create financial risk?

Portfolio questions

- How concentrated is gross profit across products?
- How many products are responsible for at least 80% of gross profit?
- How concentrated is revenue?
- Does the business depend disproportionately on a small number of products?

Geographic / operational questions

- Which regions generate the greatest gross profit?
- Which states contribute significant revenue?

- Are there states that warrant additional operational investigation based on order volume and lead-time indicators?

4. Project Objectives

The primary objective of this project is to develop a comprehensive product profitability analysis that supports data-driven commercial and operational decision-making.

The specific objectives are:

1. **Clean and validate the transactional dataset** to ensure reliable profitability calculations.
2. **Calculate key profitability metrics**, including gross margin, profit per unit and cost-to-sales ratio.
3. **Rank products by gross profit and gross margin** to distinguish financial contribution from margin efficiency.
4. **Identify high-sales/high-margin and high-sales/low-margin products** using product segmentation.
5. **Evaluate division-level financial performance** using revenue, gross profit and weighted gross margin.
6. **Analyze the relationship between sales and manufacturing cost** using correlation and scatter analysis.
7. **Identify margin and cost risks** using threshold-based product classification.
8. **Perform Pareto analysis** to determine the concentration of revenue and profit across products.
9. **Analyze regional and state-level financial concentration.**
10. **Identify products requiring business review**, including repricing, cost renegotiation and portfolio rationalization.
11. **Develop an interactive Streamlit dashboard** that enables stakeholders to explore the results dynamically.

5. Dataset Description

The analysis uses a cleaned Nassau Candy transactional dataset containing order, customer, product, geographic and financial information.

The principal fields used in the analysis include:

Field	Description
Row ID	Unique row identifier
Order ID	Unique order identifier
Order Date	Date on which the order was placed

Ship Date	Date on which the order was shipped
Ship Mode	Shipping method
Customer ID	Unique customer identifier
Country/Region	Customer country or region
City	Customer city
State/Province	Customer state or province
Postal Code	Postal or ZIP code
Division	Product division
Region	Customer region
Product ID	Unique product identifier
Product Name	Product name
Sales	Total sales value
Units	Quantity sold
Gross Profit	Sales minus manufacturing cost
Cost	Manufacturing cost
Gross Margin %	Gross profit as a percentage of sales
Profit per Unit	Gross profit divided by units sold
Cost-to-Sales Ratio	Manufacturing cost divided by sales

The final dashboard recognizes **15 distinct products** in the analyzed portfolio.

Key Business Metrics

The project uses the following primary metrics.

6.1 Gross Profit

$$[\text{Gross Profit} = \text{Sales} - \text{Cost}]$$

Gross profit represents the monetary contribution generated after deducting the manufacturing cost from sales revenue.

6.2 Gross Margin

$$[\text{Gross Margin (\%)} = \{\text{Gross Profit}\} \{\text{Sales}\} * \text{Times} 100]$$

Gross margin measures the percentage of revenue retained after manufacturing cost.

6.3 Weighted Gross Margin

For overall and division-level management reporting:

$$[\text{Weighted Gross Margin (\%)} = \{\text{sum Gross Profit}\} \{\text{sum Sales}\} * 100]$$

This measure is used instead of simply averaging individual product margins because it accounts for the relative revenue contribution of products.

The final dashboard reports an overall weighted gross margin of **65.91%**.

6.4 Profit per Unit

[Profit per Unit = {Gross Profit} / {Units}]

This metric identifies how much gross profit is generated, on average, from each unit sold.

6.5 Cost-to-Sales Ratio

[Cost to Sales = {Cost} / {Sales}]

A higher cost-to-sales ratio indicates that a greater proportion of revenue is consumed by manufacturing cost.

6.6 Profit Contribution

[Profit Contribution (%) = {Product Gross Profit} / {Total Gross Profit} * 100]

This metric identifies how much each product contributes to total portfolio gross profit.

7. Project Outcome

The project produces two major outputs:

Analytical Output

A structured profitability analysis covering:

- Product profitability
- Product margin
- Profit contribution
- Division performance
- Cost efficiency
- Risk classification
- Pareto concentration
- Regional performance
- State-level operational screening

Technology Output

An interactive **Streamlit web application** that enables users to filter the analysis by:

- Order date
- Division
- Minimum gross margin
- Product name

and download analytical results for further business use.

7. Data Loading and Understanding

The first stage of the Nassau Candy Product Line Profitability Analysis was to load the transactional dataset into Python and understand its structure, data types, variables, and business meaning. The dataset contains product-level sales transactions along with information about customers, products, divisions, locations, units, sales, manufacturing cost, and gross profit.

The dataset was loaded into a Pandas DataFrame using Python. Pandas was selected because it provides efficient functionality for data loading, inspection, cleaning, transformation, aggregation, and statistical analysis.

7.1 Data Loading

The dataset was loaded using:

```
import pandas as pd

df = pd.read_csv("Nassau_Candy_Cleaned.csv")

df.head()
```

The head () function was used to inspect the first few records and verify that the dataset had been loaded correctly.

The dataset contains the following major groups of variables:

- Transaction information: row_id, order_id, order_date, ship_date, ship_mode
- Customer information: customer_id, country/region, city, state/province, postal_code
- Product information: division, region, product_id, product_name
- Financial information: sales, cost, gross_profit
- Quantity information: units
- Derived profitability metrics: gross_margin_%, profit_per_unit, cost_to_sales_ratio

7.2 Dataset Structure

The dataset contains 20 fields in its initial cleaned form, including the original transactional and financial variables as well as calculated profitability fields.

The principal financial relationship in the dataset is:

For example, a transaction with sales of \$10.80 and cost of \$3.30 produces a gross profit of \$7.50.

This relationship was also validated during the data preparation stage.

7.3 Understanding the Business Variables

Before performing profitability analysis, each important variable was examined according to its business purpose.

Variable	Business Purpose
Order ID	Identifies the customer order
Order Date	Used for time-based analysis and filtering
Ship Date	Used to derive shipping lead time
Division	Used to compare product divisions
Region	Used for geographic profitability analysis
Product ID	Identifies individual products
Product Name	Used for product-level profitability analysis
Sales	Measures revenue generated
Units	Measures sales volume
Gross Profit	Measures profit after manufacturing cost
Cost	Measures manufacturing expenditure
Gross Margin %	Measures profitability efficiency
Profit per Unit	Measures profit generated per unit
Cost-to-Sales Ratio	Measures the proportion of revenue consumed by cost

The initial data-loading and understanding stage established that the dataset contained the required information to perform a comprehensive profitability analysis. In particular, the availability of Sales, Cost, Gross Profit, Units, Product, Division and geographic fields allowed the project to move beyond simple revenue analysis and evaluate actual financial contribution.

The dataset was subsequently subjected to data cleaning and validation before calculating the final profitability metrics and developing the Streamlit dashboard.

8. Analytical Methodology

8.1 Data Cleaning & Validation

Data cleaning and validation was performed to ensure that the dataset used for the profitability analysis was accurate, consistent, and suitable for financial analysis. NO duplicate record found and no missing values found. Since the primary focus of the project is to evaluate product profitability, particular attention was given to the **Sales, Cost, Units, and Gross Profit** fields. The objective of this stage was to identify invalid records, handle missing values appropriately, standardize categorical information, and verify the consistency of the financial data before proceeding with further analysis. No missing values found and no duplicate record found.

The **Sales and Cost** fields were examined to identify missing, invalid, zero, or negative values. Records with zero sales or invalid financial values were excluded from the profitability analysis because they could distort gross margin, profit contribution, and cost-efficiency calculations. Cost values were also validated to ensure that they represented meaningful manufacturing costs and were not negative.

The **Gross Profit** field was validated using the underlying financial relationship between sales and manufacturing cost. Gross profit is defined as the difference between sales and cost. The calculated values were compared with the existing gross profit values in the dataset. The validation showed that the values were consistent, apart from extremely small floating-point differences caused by numerical precision. These negligible differences were not considered actual business discrepancies.

The **Units** field was also reviewed because it is required for calculating profit per unit. Missing unit values were handled carefully rather than automatically being treated as zero. This distinction is important because a missing quantity does not necessarily indicate that no units were sold. Incorrectly replacing missing values with zero could result in misleading profit-per-unit calculations and affect product-level comparisons.

Product and division labels were standardized to ensure consistency during grouping and aggregation. Leading and trailing spaces and minor formatting inconsistencies were removed from product names, divisions, regions, and state/province fields. This ensured that identical products or divisions were not incorrectly treated as separate categories during the profitability and performance analysis.

The **Order Date and Ship Date** fields were validated and converted into a consistent date format to support date-based analysis and the calculation of shipment lead time. The original dates from the source dataset were preserved without artificially modifying, clipping, or replacing them. The difference between the shipment date and order date was subsequently used as a lead-time indicator for the operational analysis.

Additional validation was performed on the derived profitability measures, including **Gross Margin, Profit per Unit, and Cost-to-Sales Ratio**. These metrics were recalculated using the validated Sales, Cost, Gross Profit, and Units values to ensure consistency throughout the analysis. This helped establish a reliable foundation for product-level profitability ranking, division-level performance analysis, cost diagnostics, and Pareto analysis.

Overall, the data cleaning and validation process resulted in a more reliable analytical dataset while preserving the original business information. The cleaned data was then used for **feature engineering and profitability analysis**, ensuring that the subsequent business insights and recommendations were based on consistent and validated financial measures.

8.2 Profitability Metric Calculation

For each product, three key profitability metrics were calculated: Gross Margin (%), Profit per Unit, and Total Profit Contribution.

- **Gross Margin (%):** Measures the percentage of sales retained as gross profit after manufacturing costs. The overall average gross margin was 66.51%. At the product level, Wonka Bar - Nutty Crunch Surprise recorded a margin of 71.35%, while Wonka Bar - Milk Chocolate recorded 64.92%.
- **Profit per Unit:** Measures gross profit generated from each unit sold. Lickable Wallpaper had the highest profit per unit at \$10.00, followed by Everlasting Gobstopper at \$8.00 and Hair Toffee at \$3.50. Among the high-volume products, Wonka Bar - Scrumdiddlyumptious generated \$2.50 per unit.
- **Total Profit Contribution:** Measures each product's absolute contribution to total gross profit. Wonka Bar - Scrumdiddlyumptious was the largest contributor with \$19,357.50, followed by Wonka Bar - Triple Dazzle Caramel (\$18,610.20) and Wonka Bar - Milk Chocolate (\$17,443.37).

Key Insight: The analysis shows that the product with the highest margin or profit per unit is not necessarily the largest contributor to total profit. Therefore, margin efficiency, unit economics, and sales volume must be evaluated together when making pricing, cost optimization, and portfolio decisions.

8.3 Product-Level Profitability Analysis

Product-level profitability analysis was performed to rank products based on **total gross profit and gross margin**. The objective was to distinguish products that generate the greatest absolute profit from those that provide the highest profitability efficiency.

Gross Profit Ranking

Based on total gross profit, **Wonka Bar - Scrumdiddlyumptious** ranked first with **\$19,357.50**, contributing **20.72%** of total gross profit. It was followed by **Wonka Bar - Triple Dazzle Caramel** at **\$18,610.20 (19.92%)** and **Wonka Bar - Milk Chocolate** at **\$17,443.37 (18.67%)**.

The next two major contributors were **Wonka Bar - Nutty Crunch Surprise**, with **\$16,819.95 (18.00%)**, and **Wonka Bar - Fudge Mallows**, with **\$16,593.60 (17.76%)**. Together, these five products account for the majority of the portfolio's gross profit, demonstrating a high concentration of profitability among the major Wonka Bar products.

At the lower end, products such as **Fun Dip (\$4.80)**, **Nerds (\$7.00)**, **Sweetarts (\$28.70)**, and **Laffy Taffy (\$33.48)** make only a very small contribution to total gross profit.

Gross Margin Ranking

Gross margin ranking provides a different perspective by measuring profitability relative to sales rather than absolute profit contribution. The analysis showed that **Wonka Bar - Nutty Crunch Surprise** achieved a gross margin of **71.35%**, followed by **Wonka Bar - Scrumdiddlyumptious** at **69.44%**. **Wonka Bar - Triple Dazzle Caramel** recorded **65.33%**, while **Wonka Bar - Milk Chocolate** recorded **64.92%**.

The comparison demonstrates that the product with the highest gross margin is not necessarily the product with the highest total gross profit. For example, **Wonka Bar - Nutty Crunch Surprise** has a higher margin of 71.35%, whereas **Wonka Bar - Scrumdiddlyumptious** generates the highest absolute gross profit of \$19,357.50.

Key Business Insight

The analysis indicates that **Wonka Bar - Scrumdiddlyumptious is the strongest product in terms of absolute profit contribution**, while **Wonka Bar - Nutty Crunch Surprise demonstrates stronger margin efficiency**. Therefore, management should evaluate both **gross profit and gross margin** when making pricing, production, and portfolio decisions rather than relying on sales volume or a single profitability metric.

The relatively small profit contribution from several lower-performing products also provides an opportunity for further analysis to determine whether these products require **repricing, cost optimization, increased sales support, or portfolio rationalization review**.

8.3.1 High-Profit / High-Margin Products

The product portfolio was evaluated using **total gross profit and gross margin** to identify products that combine strong absolute profitability with efficient margin performance. This analysis is important because a product with a high gross margin does not necessarily generate the highest total profit, particularly when its sales volume is relatively low.

The analysis identifies **Wonka Bar - Scrumdiddlyumptious** as the strongest product in terms of total gross profit, generating **\$19,357.50** and contributing **20.72%** of total gross profit. It also maintains a strong gross margin of **69.44%** and generates **\$2.50 profit per unit** across **7,743 units** sold.

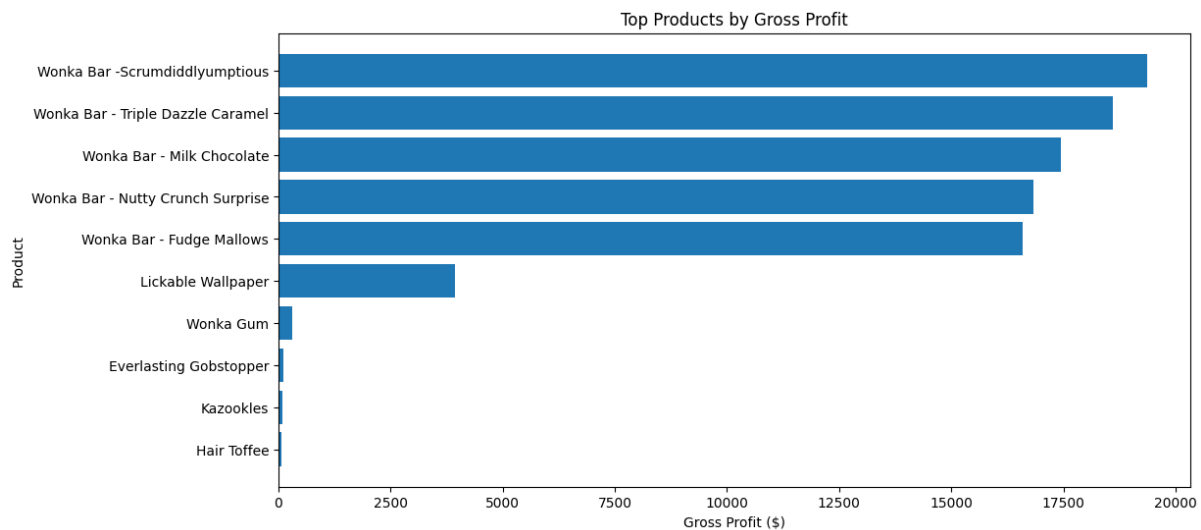
Wonka Bar - Nutty Crunch Surprise demonstrates the strongest margin efficiency among the major high-profit products, with a gross margin of **71.35%**. It generated **\$16,819.95** in gross profit, contributed **18.00%** of total profit, and generated **\$2.49 profit per unit** from **6,755 units**.

Other major high-profit products include **Wonka Bar - Triple Dazzle Caramel**, which generated **\$18,610.20** in gross profit with a **65.33% gross margin**, and **Wonka Bar - Milk Chocolate**, which generated **\$17,443.37** with a **64.92% gross margin**. **Wonka Bar - Fudge Mallows** also made a significant contribution of **\$16,593.60** and generated **\$2.40 profit per unit**.

An important finding is that **Lickable Wallpaper** has the highest profit per unit at **\$10.00**, but its total contribution is lower at **\$3,930.00** because only **393 units** were sold. This demonstrates that high unit profitability does not necessarily translate into high overall profit contribution.

Overall, the analysis indicates that the major **Wonka Bar products are the core profit drivers**, combining high sales volume, substantial gross profit, and relatively strong margins. These products should therefore be prioritized for **inventory availability, pricing protection, production planning, and growth initiatives**. At the same time, high-margin but lower-

volume products such as Lickable Wallpaper may represent opportunities for targeted sales and marketing efforts to increase their overall contribution.



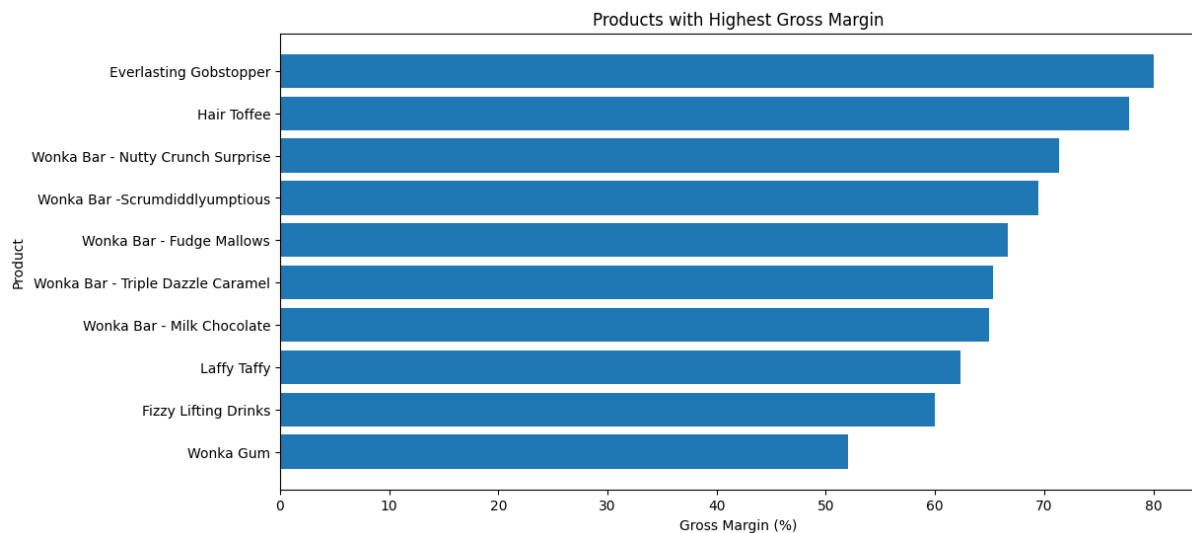
8.3.2 Gross Margin Analysis

Gross margin analysis was performed to identify products that generate the highest profitability relative to their sales revenue. The chart shows considerable variation in gross margin across the product portfolio, indicating differences in pricing and cost efficiency.

Everlasting Gobstopper has the highest gross margin at approximately **80%**, followed by **Hair Toffee** at approximately **78%**. Among the major high-volume products, **Wonka Bar - Nutty Crunch Surprise** records a strong gross margin of **71.35%**, followed by **Wonka Bar - Scrumdiddlyumptious** at **69.44%**.

Other major Wonka products show relatively strong margins, including **Wonka Bar - Fudge Mallows** at approximately **66%**, **Wonka Bar - Triple Dazzle Caramel** at **65.33%**, and **Wonka Bar - Milk Chocolate** at **64.92%**. **Laffy Taffy** and **Fizzy Lifting Drinks** also demonstrate relatively healthy margins of approximately **62%** and **60%**, respectively.

At the lower end of the chart, **Wonka Gum** has the lowest gross margin among the displayed products at approximately **52%**. Although this is still a positive margin, it is considerably lower than the margins achieved by the leading products.



Key Insight

The analysis shows that **high gross margin and high total profit contribution are not necessarily the same**. For example, Everlasting Gobstopper has the highest margin, but its total profit contribution is only **\$104**, whereas Wonka Bar - Scrumdiddlyumptious has a lower margin of **69.44%** but generates the highest total gross profit of **\$19,357.50** because of its substantially higher sales volume.

Therefore, management should evaluate **gross margin together with sales volume, total gross profit, and profit per unit**. Products with high margins but low volumes may represent **growth opportunities**, while high-volume products with comparatively lower margins should be monitored for **pricing and cost optimization**.

8.3.3 Product Profitability Segmentation

To obtain a more actionable view of the product portfolio, products were segmented using **sales volume, gross profit, and gross margin**. This classification helps management distinguish products that should be protected and expanded from products that require pricing, cost, or portfolio review.

High-Profit / High-Margin Products

The analysis identified five products that combine high total gross profit with relatively strong gross margins. These products are important contributors to the overall profitability of the portfolio.

Product	Sales (\$)	Gross Profit (\$)	Gross Margin (%)	Profit/Unit (\$)
Wonka Bar - Fudge Mallows	24,890.40	16,593.60	66.67	2.40
Wonka Bar - Milk Chocolate	26,867.75	17,443.37	64.92	2.11
Wonka Bar - Nutty Crunch Surprise	23,574.95	16,819.95	71.35	2.49
Wonka Bar - Triple Dazzle Caramel	28,485.00	18,610.20	65.33	2.45
Wonka Bar Scrumdiddlyumptious	27,874.80	19,357.50	69.44	2.50

Wonka Bar - Scrumdiddlyumptious generates the highest gross profit at \$19,357.50, with a strong gross margin of 69.44% and the highest profit per unit among these five products at \$2.50.

Wonka Bar - Nutty Crunch Surprise records the highest gross margin at 71.35%, while generating \$16,819.95 in gross profit and \$2.49 profit per unit. This indicates strong margin efficiency combined with substantial profit contribution.

Wonka Bar - Triple Dazzle Caramel generates the highest sales among these products at \$28,485.00 and contributes \$18,610.20 in gross profit, with a 65.33% gross margin.

Wonka Bar - Milk Chocolate also makes a substantial contribution, generating \$17,443.37 in gross profit from \$26,867.75 in sales. Its gross margin of 64.92% is the lowest among the five products, but its high sales volume makes it an important profit contributor.

Key Insight

These five products demonstrate a combination of scale, profitability, and positive margin performance. Together, they represent the core group of high-profit products in the portfolio. Among them, Scrumdiddlyumptious stands out for total gross profit, while Nutty Crunch Surprise stands out for gross-margin efficiency.

From a business perspective, these products warrant close monitoring of pricing, inventory availability, manufacturing costs, and supply continuity, because changes in their sales or margins could have a meaningful effect on overall portfolio profitability.

Business implication: These products should be prioritized for inventory availability, production planning, pricing protection, and growth initiatives.

High-Sales / Low-Margin Products

For the report, this table should identify products that generate **high sales but relatively lower gross profit or margin**. These products are important because high sales volume may create the appearance of strong performance while profitability remains comparatively weak.

Product Name	Total Sales	Total Cost	Total Gross Profit	Total Units	Average Gross Margin	Average Profit per Unit
Wonka Bar - Milk Chocolate	\$26,867.75	\$9,424.38	\$17,443.37	8,267	64.92%	\$2.11
Wonka Bar - Triple Dazzle Caramel	\$28,485.00	\$9,874.80	\$18,610.20	7,596	65.33%	\$2.45
Wonka Bar - Fudge Mallows	\$24,890.40	\$8,296.80	\$16,593.60	6,914	66.67%	\$2.40

Report Interpretation

The analysis shows that **Milk Chocolate** has high sales of **\$26,867.75** and the highest unit volume among these products at **8,267 units**, but its average profit per unit is only **\$2.11** and gross margin is **64.92%**. This indicates that strong sales volume does not necessarily translate into proportionately strong unit profitability.

Triple Dazzle Caramel generates the highest sales in this group at **\$28,485.00** and gross profit of **\$18,610.20**, but its **65.33% gross margin** remains below the higher-margin products such as Nutty Crunch Surprise. This makes it a relevant product for monitoring pricing and cost efficiency.

Fudge Mallows records **\$24,890.40 in sales** and **\$16,593.60 in gross profit**, with a **66.67% margin**. Although profitability is relatively strong, its profit contribution is lower than the leading products such as Scrumdiddlyumptious.

Business implication: High-sales products should not be evaluated solely on revenue. Management should also monitor **gross margin, profit per unit, and total profit contribution** to identify opportunities for pricing optimization, cost control, and product-mix improvement.

Low-Sales / Low-Profit Products

The analysis identifies a group of products with relatively low sales volumes and limited total gross-profit contribution. However, the results also show that **low sales do not always mean poor profitability**. Some products generate high profit per unit despite low overall sales, while others combine low sales with very weak margins.

Product	Total Sales	Total Cost	Total Gross Profit	Units	Avg. Gross Margin	Profit/Unit
Everlasting Gobstopper	\$130.00	\$26.00	\$104.00	13	80.00%	\$8.00
Fizzy Lifting Drinks	\$78.75	\$31.50	\$47.25	21	60.00%	\$2.25
Fun Dip	\$12.00	\$7.20	\$4.80	8	40.00%	\$0.60
Hair Toffee	\$76.50	\$17.00	\$59.50	17	77.78%	\$3.50
Kazookles	\$1,205.75	\$1,113.00	\$92.75	371	7.69%	\$0.25
Laffy Taffy	\$53.73	\$20.25	\$33.48	27	62.31%	\$1.24
Lickable Wallpaper	\$7,860.00	\$3,930.00	\$3,930.00	393	50.00%	\$10.00
Nerds	\$15.00	\$8.00	\$7.00	10	46.67%	\$0.70
Sweetarts	\$61.50	\$32.80	\$28.70	41	46.67%	\$0.70
Wonka Gum	\$597.50	\$286.80	\$310.70	478	52.00%	\$0.65

Key Findings

1. Kazookles represents the clearest low-profit concern.

It generates **\$1,205.75 in sales across 371 units**, but produces only **\$92.75 in gross profit**. Its **7.69% gross margin** and **\$0.25 profit per unit** indicate that a large proportion of sales revenue is absorbed by cost.

2. Wonka Gum has relatively high unit volume but limited profitability.

With **478 units sold**, Wonka Gum has the highest unit volume in this group. However, it generates only **\$310.70 gross profit** and **\$0.65 profit per unit**, with a **52% margin**. This suggests that sales volume alone is not sufficient to make the product a major profit contributor.

3. Lickable Wallpaper is an important exception.

Although its sales volume is relatively modest at **393 units**, it generates **\$3,930 gross profit** and the highest **\$10.00 profit per unit** in the group. Therefore, it should not be classified as a poor-performing product simply because its sales volume is lower.

4. Everlasting Gobstopper and Hair Toffee demonstrate strong unit economics.

Everlasting Gobstopper achieves an **80% gross margin** and **\$8.00 profit per unit**, while Hair Toffee achieves **77.78% margin** and **\$3.50 profit per unit**. Their low sales volumes limit their overall contribution, but their underlying margins are strong.

Business Interpretation

The analysis demonstrates that **sales volume and profitability must be evaluated together**. Products such as **Kazookles and Wonka Gum** require closer cost and pricing review because their relatively higher unit volumes do not translate into substantial profit. In contrast, **Lickable Wallpaper, Everlasting Gobstopper, and Hair Toffee** have attractive unit economics despite lower sales volumes.

From a management perspective, the appropriate action should therefore differ by product: **low-margin products require cost or pricing analysis**, while **high-margin, low-volume products may represent opportunities for targeted sales growth**.

Overall Finding

The segmentation demonstrates why **sales volume alone is not an adequate measure of product success**. The most valuable products are those that combine strong volume, healthy margins, and substantial absolute profit, while high-volume products with comparatively lower margins provide opportunities for margin improvement. Low-volume, low-profit products should be reviewed to determine whether they have sufficient strategic value to justify their continued presence in the portfolio.

9. Division-Level Performance Analysis

9.1 Aggregate Matrices by Division

The division-level aggregate matrix summarizes sales, cost, gross profit, volume, and profitability efficiency across the three business divisions.

Division	Total Sales	Total Cost	Total Gross Profit	Total Units	Avg. Gross Margin	Avg. Profit / Unit
Chocolate	\$131,692.90	\$42,868.28	\$88,824.62	37,275	67.46%	\$2.38
Other	\$9,663.25	\$5,329.80	\$4,333.45	1,242	37.67%	\$3.36
Sugar	\$427.48	\$142.75	\$284.73	137	57.69%	\$1.89

Division-Level Analysis

Chocolate is the largest division by a substantial margin, generating **\$131,692.90 in sales** and **\$88,824.62 in gross profit** from **37,275 units**. Its **67.46% average gross margin** indicates strong profitability at scale. The division therefore represents the primary contributor to the overall business profitability.

The **Other** division generates **\$9,663.25 in sales** and **\$4,333.45 in gross profit**. Although its gross margin of **37.67%** is considerably lower than Chocolate and Sugar, it has the **highest average profit per unit at \$3.36**. This indicates that its profitability is supported by higher profit generated per individual unit despite a weaker overall margin percentage.

The **Sugar** division has the smallest scale, with **\$427.48 in sales**, **137 units**, and **\$284.73 in gross profit**. Its **57.69% gross margin** indicates reasonable margin performance, but its limited sales volume means that its contribution to total profitability remains small.

Key Business Insights

- **Chocolate:** High-scale and high-margin division, with **67.46% margin** and **\$88,824.62 gross profit**.

- **Other:** Lowest gross margin (**37.67%**) but highest profit per unit (**\$3.36**), indicating a different profitability structure.
- **Sugar:** Moderate margin (**57.69%**) but very low sales volume, limiting its overall contribution.
- The analysis shows why division performance should be evaluated using **both margin percentage and absolute profit**, rather than relying on a single metric.

9.2 Comparison of Average Gross Margin by Division

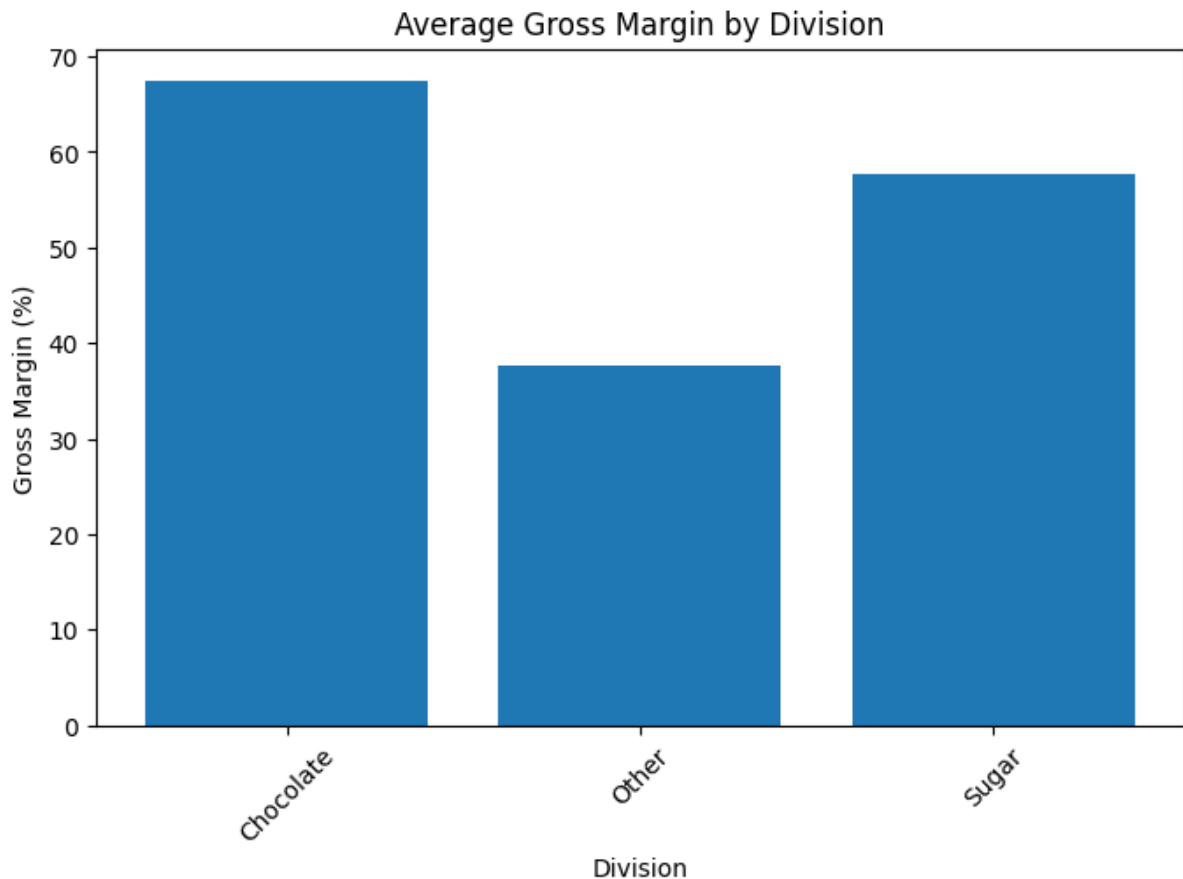
Division	Average Gross Margin	Profit Contribution	Average Profit / Unit
Chocolate	67.46%	95.06%	\$2.38
Sugar	57.69%	0.30%	\$1.89
Other	37.67%	4.64%	\$3.36

Analysis

- The **Chocolate division** records the highest average gross margin at **67.46%**, indicating strong margin efficiency. It also contributes **95.06% of total profit**, making it the dominant contributor to overall profitability. Its average profit per unit is **\$2.38**.
- The **Sugar division** has an average gross margin of **57.69%**, which is **9.77 percentage points lower than Chocolate**. However, its profit contribution is only **0.30%**, primarily because of its very small sales and unit volume.
- The **Other division** has the **lowest average gross margin at 37.67%**, which is **29.79 percentage points below Chocolate** and **20.02 percentage points below Sugar**. Despite the lower margin, it generates the **highest average profit per unit at \$3.36**. Its profit contribution is **4.64%**.

Key Insight

- The comparison highlights that **gross margin percentage and profit per unit measure different aspects of profitability**. Chocolate demonstrates the strongest overall margin performance and dominates total profit contribution, while Other generates more profit per unit despite having the lowest average margin. Sugar maintains a relatively healthy margin but has limited business scale.
- **Management implication:** The **Other division** warrants closer review of its cost structure and pricing because of its low **37.67% margin**, while the **Chocolate division's** strong margin and dominant contribution make maintaining its cost and pricing efficiency important for overall profitability.



9.2.2 Revenue vs. Profit Imbalance by Division

The comparison between revenue and gross profit highlights how effectively each division converts sales into profit.

Division	Total Sales	Total Gross Profit	Profit as % of Sales
Chocolate	\$131,692.90	\$88,824.62	67.46%
Other	\$9,663.25	\$4,333.45	44.85%
Sugar	\$427.48	\$284.73	66.61%

Analysis

The **Chocolate division** generates by far the highest revenue at **\$131,692.90** and also produces **\$88,824.62 in gross profit**. Approximately **67.46% of its sales translates into gross profit**, indicating strong revenue-to-profit conversion. Its large revenue base makes it the principal driver of overall profitability.

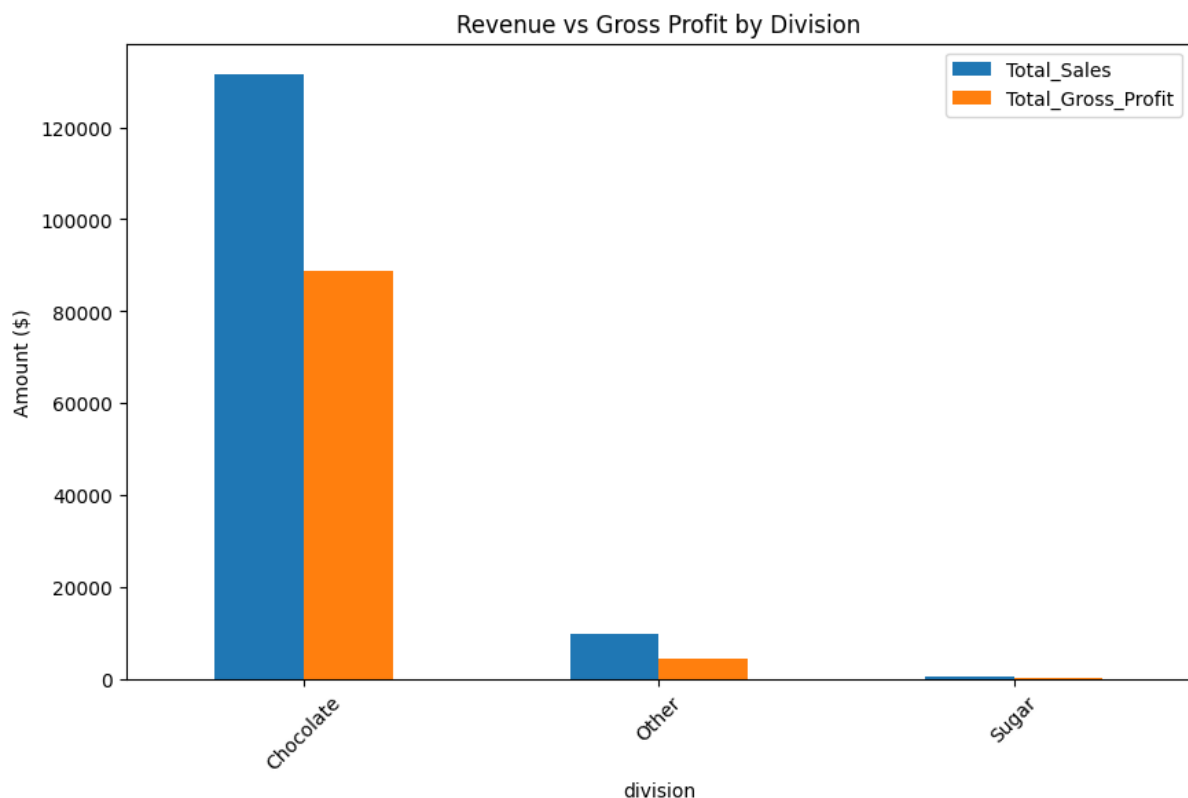
The **Other division** generates **\$9,663.25 in sales** but only **\$4,333.45 in gross profit**, giving a profit-to-sales ratio of approximately **44.85%**. This represents the largest revenue-to-profit imbalance among the three divisions, with more than half of its sales absorbed by costs.

The **Sugar division** has very small revenue of **\$427.48**, but generates **\$284.73 in gross profit**, corresponding to approximately **66.61% of sales**. Although its revenue contribution is minimal, its revenue-to-profit conversion is relatively strong.

Key Insight

The analysis reveals that **high revenue does not automatically indicate proportional profitability**. Chocolate combines high revenue with strong profit conversion, whereas Other generates substantially less profit relative to its revenue. Sugar demonstrates strong conversion efficiency but operates at a very small scale.

Business implication: Revenue should therefore be assessed alongside gross profit and profit-to-sales conversion. The **Other division** presents the clearest opportunity for investigating cost structure and pricing efficiency, while Chocolate's strong revenue and profit generation make its margin preservation particularly important.



The graph shows that **Chocolate** has the highest revenue (**\$131,692.90**) and gross profit (**\$88,824.62**), making it the dominant division. **Other** generates significantly lower revenue (**\$9,663.25**) and gross profit (**\$4,333.45**), while **Sugar** has the lowest revenue (**\$427.48**) and profit (**\$284.73**).

9.3 Identify divisions with:

* Strong financial efficiency:

The **Chocolate division** demonstrates strong financial efficiency, generating **\$131,692.90 in total sales** and **\$88,824.62 in gross profit** from **37,275 units**. Its **67.46% average gross**

margin indicates that a substantial proportion of revenue is retained as gross profit after accounting for product costs.

The division also contributes **95.06% of total company profit**, making it the primary driver of overall profitability. With an **average profit of \$2.38 per unit**, Chocolate combines **high sales volume, strong margins, and substantial absolute profit generation**.

Overall, the Chocolate division represents a financially efficient and highly significant business segment, with strong revenue-to-profit conversion and a dominant contribution to total profitability.

9.4 Structural Margin Issues – Division-Level Analysis

Based on the division-level profitability metrics, the **Other division shows the clearest structural margin issue**.

Division	Total Sales	Total Cost	Gross Profit	Units	Avg. Gross Margin	Avg. Profit / Unit	Profit Contribution
Chocolate	\$131,692.90	\$42,868.28	\$88,824.62	37,275	67.46%	\$2.38	95.06%
Other	\$9,663.25	\$5,329.80	\$4,333.45	1,242	37.67%	\$3.36	4.64%
Sugar	\$427.48	\$142.75	\$284.73	137	57.69%	\$1.89	0.30%

Analysis

The **Other division** has the lowest average gross margin at **37.67%**, compared with **67.46% for Chocolate** and **57.69% for Sugar**. Its total cost of **\$5,329.80** absorbs a relatively large portion of its **\$9,663.25 sales**, limiting its overall profit contribution to **4.64%**.

This indicates a potential **structural margin issue**, where the division's cost structure or pricing model may be limiting profitability. Interestingly, its **\$3.36 average profit per unit is the highest** among the divisions, suggesting that the low margin percentage should be investigated alongside product mix, pricing, and unit economics rather than viewed in isolation.

Business implication: The Other division should be prioritized for a deeper **cost and pricing review**, particularly to identify products or cost components responsible for the comparatively low **37.67% margin**.

10. Profit Concentration (Pareto) Analysis

- **Determine % of products contributing:**

- **80% of revenue**

- 80% of profit

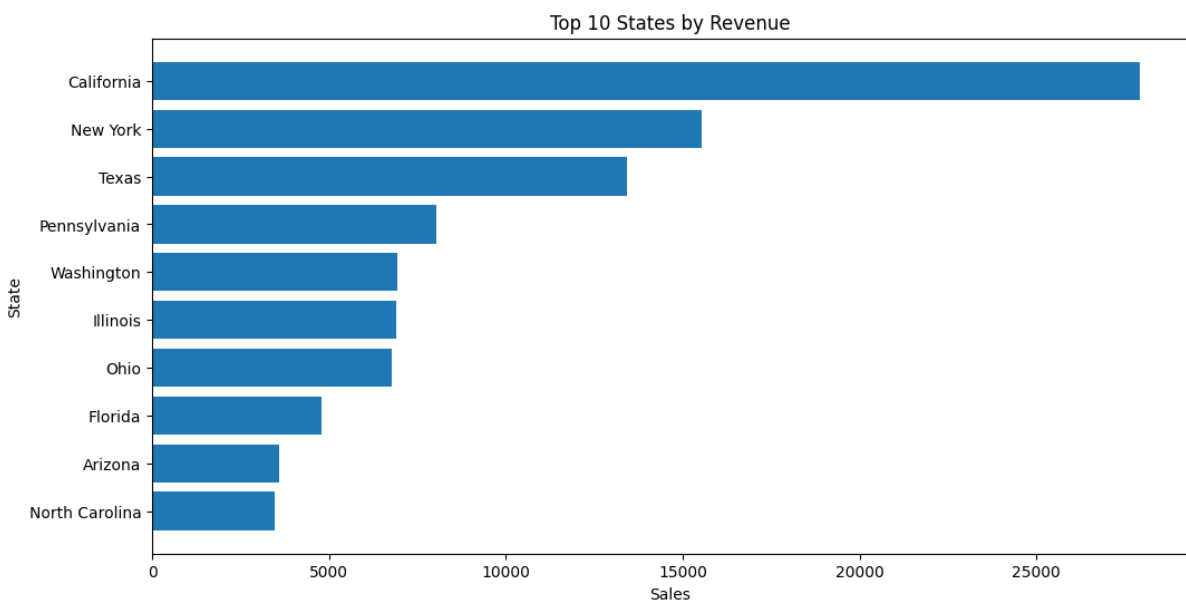
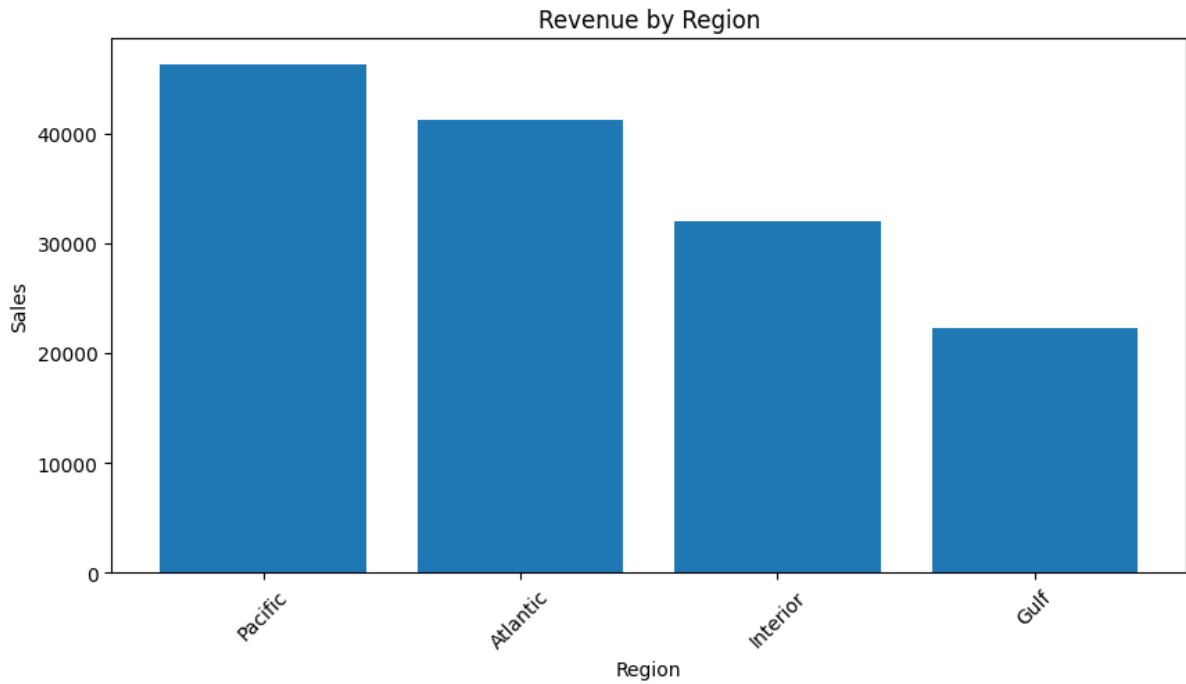
10.1 Products Contributing 80% of Revenue

The revenue analysis shows that the top four products generate 76.26% of total revenue:

Rank	Product	Revenue	Revenue %	Cumulative Revenue %
1	Wonka Bar - Triple Dazzle Caramel	\$28,485.00	20.09%	20.09%
2	Wonka Bar Scrumdiddlyumptious	\$27,874.80	19.66%	39.75%
3	Wonka Bar - Milk Chocolate	\$26,867.75	18.95%	58.70%
4	Wonka Bar - Fudge Mallows	\$24,890.40	17.56%	76.26%

The 80% revenue threshold is therefore not reached within the four products shown. The next product, Wonka Bar - Nutty Crunch Surprise, would be required to cross the 80% threshold.

Interpretation: Revenue is highly concentrated among a small group of Wonka Bar products, indicating that these products are important contributors to overall sales performance.



10.2 Products Contributing 80% of Profit

The profit analysis shows:

Rank	Product	Total Profit	Profit %	Cumulative Profit %
1	Wonka Bar Scrumdiddlyumptious	\$19,357.50	20.72%	20.72%

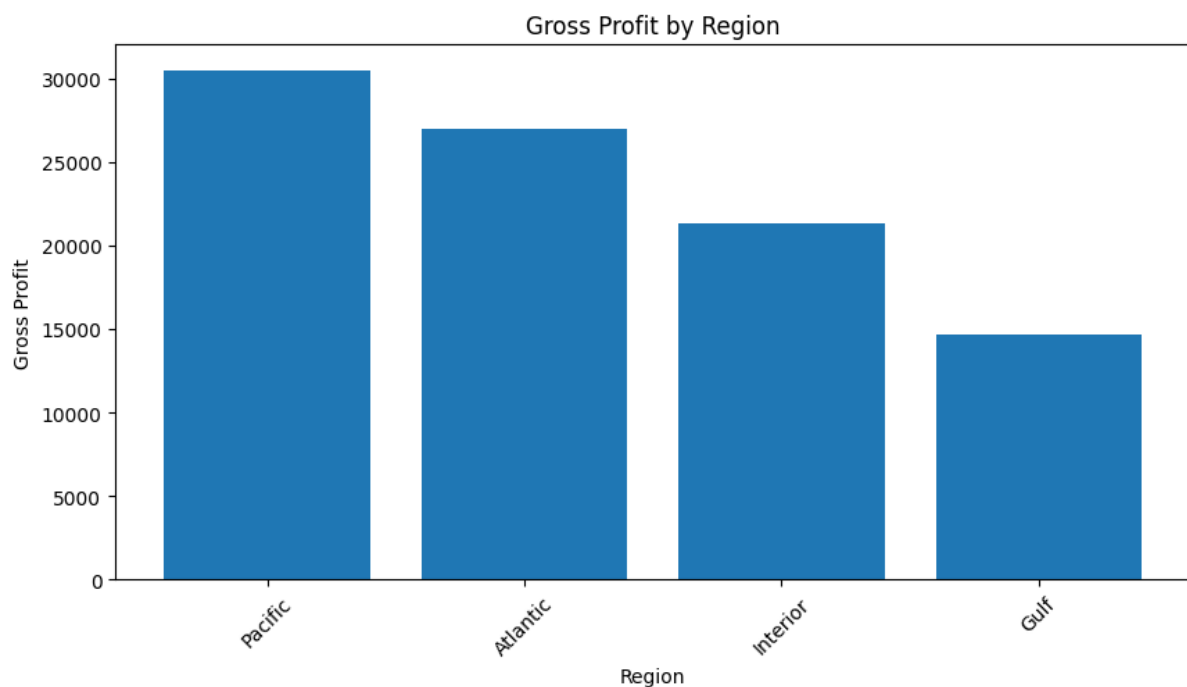
Rank	Product	Total Profit	Profit %	Cumulative Profit %
2	Wonka Bar - Triple Dazzle Caramel	\$18,610.20	19.92%	40.63%
3	Wonka Bar - Milk Chocolate	\$17,443.37	18.67%	59.30%
4	Wonka Bar - Nutty Crunch Surprise	\$16,819.95	18.00%	77.30%

The top four products contribute 77.30% of total profit. Therefore, the 80% profit threshold would be crossed by adding the next highest-profit product, Wonka Bar - Fudge Mallows.

Key Finding

The Pareto analysis demonstrates a high concentration of both revenue and profit within a small number of products. The leading Wonka Bar products account for the majority of financial contribution, indicating that product performance is concentrated rather than evenly distributed across the portfolio.

Business implication: These high-contribution products should receive close attention in areas such as inventory planning, pricing, cost management, production continuity, and margin monitoring, because changes in their performance could have a meaningful effect on overall business results.



10.2 Detect congestion-prone states or region Identify over-dependency risks

The state-level analysis identifies **California and New York** as the two largest contributors to revenue among the states shown.

State	Total Sales	Gross Profit	Revenue Contribution
California	\$27,917.40	\$18,479.42	19.69%
New York	\$15,541.03	\$10,222.44	10.96%

Analysis

California is the largest revenue-contributing state, generating **\$27,917.40 in sales** and **\$18,479.42 in gross profit**, representing **19.69% of total revenue**. Its high contribution makes it an important market for overall business performance.

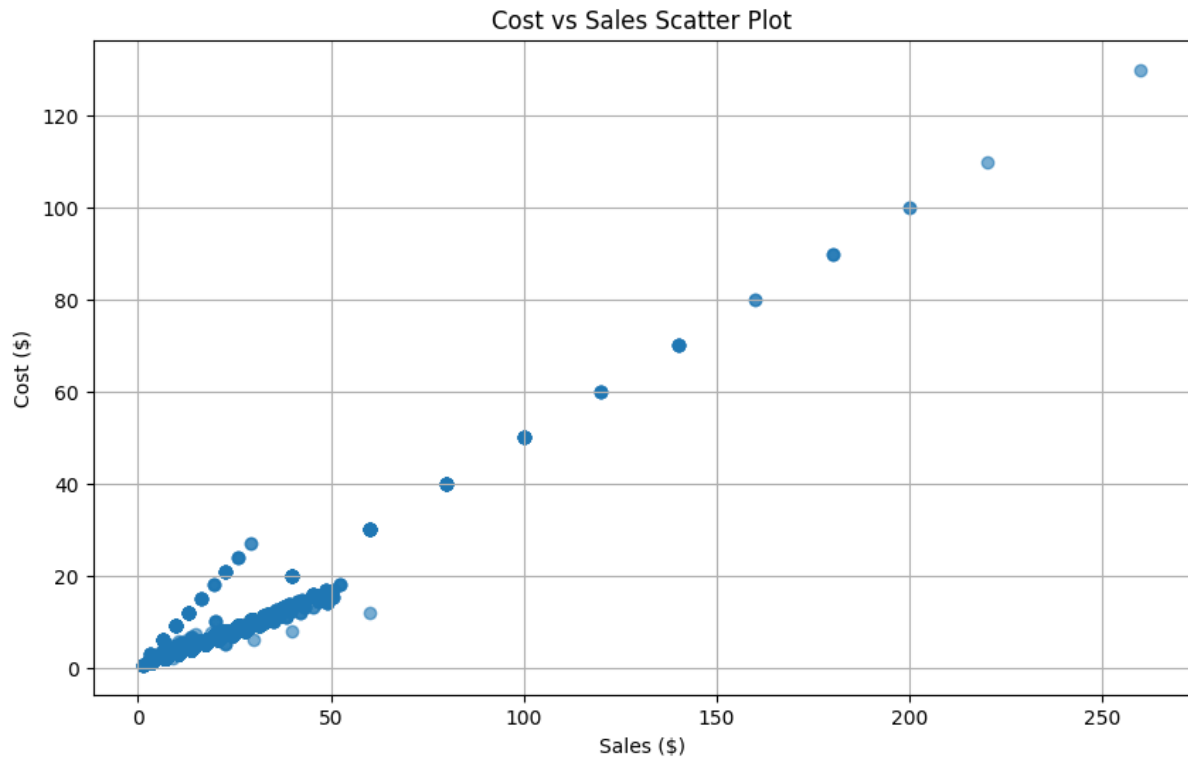
New York contributes **10.96% of total revenue**, with **\$15,541.03 in sales** and **\$10,222.44 in gross profit**. Together, California and New York account for **30.65% of total revenue**, indicating a noticeable geographic concentration.

Over-Dependency Risk

The concentration of revenue in these two states creates a potential **geographic dependency risk**. A significant change in demand, distribution efficiency, customer activity, or operating conditions in either market could affect overall revenue performance.

Business implication: The business should monitor performance and operational conditions in California and New York closely while continuing to develop revenue across other states. This would help reduce excessive dependence on a limited number of geographic markets and improve overall revenue diversification.

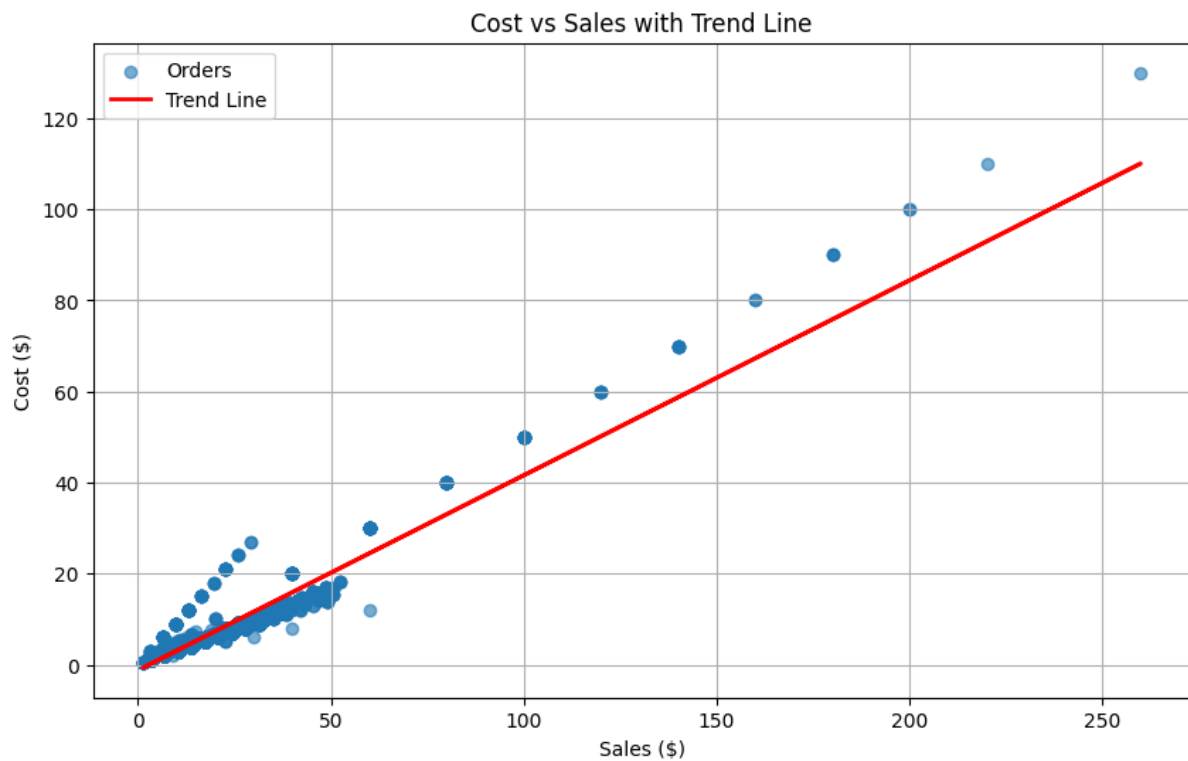
11. Cost Structure Diagnostics



The scatter plot shows a **strong positive relationship between sales and cost**. As sales increase, product cost generally increases as well, indicating that higher-revenue products tend to require higher production or procurement costs.

Most observations are concentrated in the **low-sales and low-cost range**, while a smaller number of products have substantially higher sales and corresponding costs. The overall pattern suggests that **cost is closely associated with sales volume**, although a few observations deviate from the main trend.

Business implication: The strong sales–cost relationship indicates that cost management is important for protecting profitability. Products with relatively high costs compared with their sales should be further investigated for **pricing optimization, sourcing improvements, or cost reduction opportunities**.

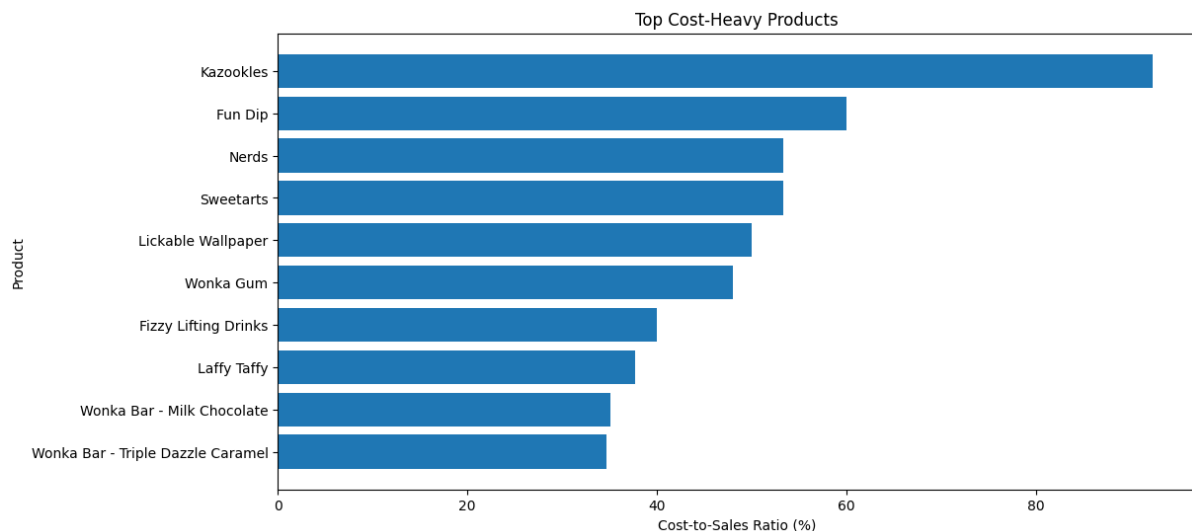


The trend line shows a **clear positive relationship between sales and cost**. As sales increase, costs generally increase as well, indicating that higher-sales products require proportionally higher costs.

Most orders are concentrated in the **lower sales and cost range**, while a few high-value observations have substantially higher sales and costs. The observations generally follow the upward trend, although some points lie above or below the trend line, indicating differences in cost efficiency across products.

Business implication: The strong upward trend highlights the importance of **cost control and pricing management**. Products positioned significantly above the trend line may require further investigation because their costs are relatively high compared with their sales. This can help identify opportunities for **cost reduction, supplier negotiation, or pricing optimization**.

Cost-to-Sales Ratio by Product



Cost-to-Sales Ratio by Product

The chart compares the **cost-to-sales ratio** across the top cost-heavy products. A higher ratio indicates that a larger proportion of sales revenue is consumed by product cost, leaving less revenue available as gross profit.

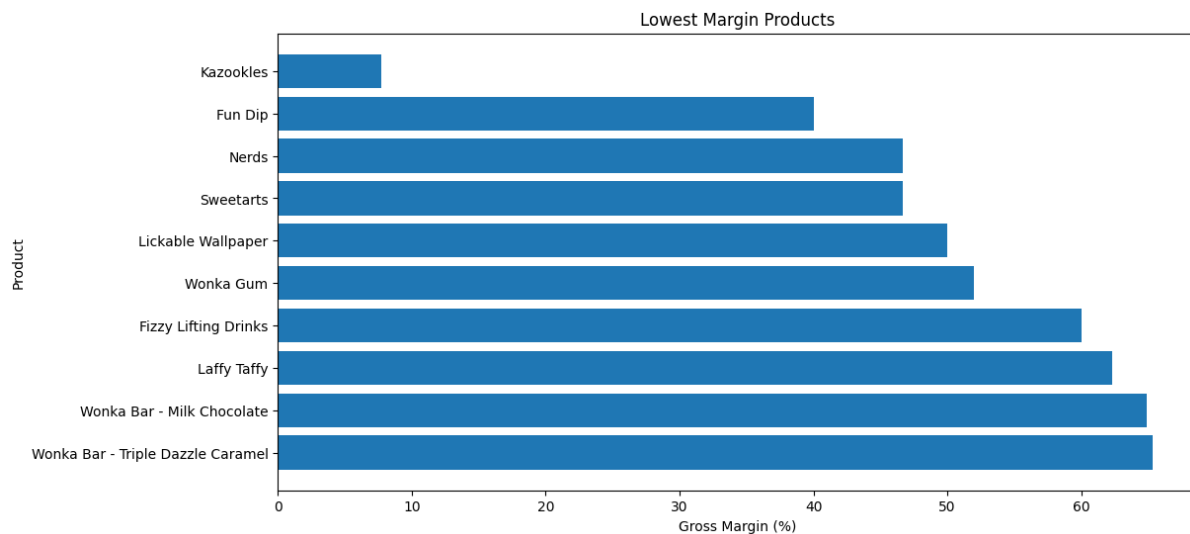
- **Kazookles** has the highest cost-to-sales ratio at approximately **92%**, indicating a significant cost burden and very limited margin.
- **Fun Dip** follows at around **60%**, showing relatively high costs compared with its sales.
- **Nerds** and **Sweetarts** are both around **53%**, indicating moderate-to-high cost pressure.
- **Lickable Wallpaper** and **Wonka Gum** are approximately **50%** and **48%**, respectively.
- **Fizzy Lifting Drinks** is around **40%**, while **Laffy Taffy** is approximately **38%**.
- **Wonka Bar – Milk Chocolate** and **Wonka Bar – Triple Dazzle Caramel** have comparatively lower ratios of around **35%**, indicating better cost efficiency among the products shown.

Key Business Insight

Kazookles is the most critical cost-efficiency concern, as approximately 92% of its sales are absorbed by cost. This is consistent with its previously observed **7.69% gross margin**. It should therefore be prioritized for **cost reduction, supplier negotiation, or pricing review**.

Overall, the chart indicates that products with a **high cost-to-sales ratio require closer margin monitoring**, while products with lower ratios have relatively greater scope to retain revenue as gross profit.

Gross Margin by Product:



The chart highlights products with the **lowest gross margins**, indicating which products retain the smallest proportion of sales as gross profit.

- **Kazookles** has the lowest gross margin at approximately **7.69%**, making it the most significant margin-risk product.
- **Fun Dip** follows with a margin of approximately **40%**.
- **Nerds** and **Sweetarts** both have margins of around **46.67%**.
- **Lickable Wallpaper** has a **50%** margin, while **Wonka Gum** records approximately **52%**.
- **Fizzy Lifting Drinks** has a **60%** margin.
- **Laffy Taffy** records approximately **62.31%**.
- Among the higher-volume products shown, **Milk Chocolate (64.92%)** and **Triple Dazzle Caramel (65.33%)** have comparatively stronger margins.

Key Business Insight

Kazookles is the most critical margin-risk product, with only **7.69% gross margin**. This is consistent with its very high cost-to-sales ratio and indicates that most of its revenue is absorbed by costs.

The analysis suggests that **Kazookles should be prioritized for pricing review, cost reduction, or supplier negotiation**, while the other low-margin products should be monitored to determine whether their margins justify their sales contribution.

Product-Level Recommendation Analysis

The product-level analysis categorizes products into **Retain** and **Discontinuation Review** based on their sales, gross profit, gross margin, and cost-to-sales ratio.

Products Recommended for Retention

The five major **Wonka Bar** products demonstrate strong financial performance and should be retained:

- **Wonka Bar – Scrumdiddlyumptious** generates the highest gross profit of **\$19,357.50**, with a **69.44% gross margin**.
- **Triple Dazzle Caramel** generates **\$18,610.20** gross profit and **65.33% margin**, with the highest sales of **\$28,485.00**.
- **Milk Chocolate** generates **\$17,443.37** gross profit from **\$26,867.75** sales, with a **64.92% margin**.
- **Nutty Crunch Surprise** has the strongest margin among the retained products at **71.35%**, generating **\$16,819.95** gross profit.
- **Fudge MalloWS** contributes **\$16,593.60** gross profit with a **66.67% margin**.

These products combine **high sales, substantial gross profit, and healthy margins**, making them strategically important to the product portfolio.

Products Requiring Discontinuation Review

The remaining products have been flagged for **discontinuation review** based on their relatively low profitability, low sales volume, or high cost-to-sales ratios.

Kazookles is the most critical concern, with a **7.69% gross margin** and an exceptionally high **92.31% cost-to-sales ratio**. Despite generating \$1,205.75 in sales, it contributes only **\$92.75 in gross profit**, indicating significant cost pressure.

Fun Dip, Nerds, and Sweetarts also show relatively weak profitability, with margins of **40.00%, 46.67%, and 46.67%**, respectively. Their low sales volumes further limit their contribution.

Wonka Gum has higher volume at **478 units**, but its **52% margin** and **\$310.70 gross profit** suggest that its overall contribution remains limited compared with the core Wonka Bar products.

Important Management Consideration

The “**Discontinuation Review**” recommendation should be treated as a **screening recommendation rather than an automatic decision to discontinue**. Products such as **Everlasting Gobstopper (80% margin)** and **Hair Toffee (77.78% margin)** have strong margins but very low sales volumes. They may therefore warrant **sales-growth or market-potential evaluation** before discontinuation.

Overall, the analysis supports a **portfolio optimization strategy**: protect and prioritize the high-performing Wonka Bar products, investigate high-cost/low-margin products such as Kazookles, and conduct a broader strategic review of low-volume products before making final discontinuation decisions.

1. Strategic Recommendation Summary

Recommendation	Products	Sales	Gross Profit	Profit Contribution
Retain	5	\$131,692.90	\$88,824.62	95.06%
Discontinuation Review	10	\$10,090.73	\$4,618.18	4.94%
Total	15	\$141,783.63	\$93,442.80	100%

The five retained products generate **92.88% of total sales and 95.06% of total gross profit**, demonstrating a highly concentrated and financially efficient core product portfolio. The remaining ten products account for only **4.94% of total profit**, supporting the need for closer portfolio review.

2. Products to Retain

The following five products should be **retained and prioritized**:

Product	Sales	Gross Profit	Margin	Units
Wonka Bar - Scrumdiddlyumptious	\$27,874.80	\$19,357.50	69.44%	7,743
Wonka Bar - Triple Dazzle Caramel	\$28,485.00	\$18,610.20	65.33%	7,596
Wonka Bar - Milk Chocolate	\$26,867.75	\$17,443.37	64.92%	8,267
Wonka Bar - Nutty Crunch Surprise	\$23,574.95	\$16,819.95	71.35%	6,755
Wonka Bar - Fudge Mallows	\$24,890.40	\$16,593.60	66.67%	6,914

These products demonstrate a strong combination of **high sales volume, high absolute gross profit, and healthy margins**. Together, they generate **\$88,824.62 in gross profit**, making them the foundation of the company's profitability.

Strategic action: Maintain inventory availability, protect pricing, monitor production costs, and prioritize these products in sales and supply-chain planning.

3. Products Requiring Review

The file identifies the following ten products for **Discontinuation Review**:

- Everlasting Gobstopper
- Fizzy Lifting Drinks
- Fun Dip
- Hair Toffee
- **Kazookles**
- Laffy Taffy
- Lickable Wallpaper
- Nerds
- Sweetarts
- Wonka Gum

However, these products should **not all be treated equally**.

High-Priority Review: Kazookles

Kazookles is the clearest margin-risk product.

-
- Sales: **\$1,205.75**
 - Cost: **\$1,113.00**
 - Gross Profit: **\$92.75**
 - Gross Margin: **7.69%**
 - Cost-to-Sales Ratio: **92.31%**
 - Units: **371**
-

Approximately **92% of its sales revenue is consumed by cost**, leaving only a small gross-profit contribution. This provides strong evidence for a detailed pricing, sourcing, cost-reduction, or discontinuation assessment.

Moderate Review: Wonka Gum

Wonka Gum generates **478 units**, but only **\$310.70 in gross profit**, with a **52% gross margin**. Its relatively high unit volume but low absolute profit contribution makes it appropriate for a portfolio-efficiency review.

Strategic Review Rather Than Immediate Discontinuation

Some products flagged for review actually have

strong margins but very low sales volume:

-
- **Everlasting Gobstopper:** 80.00% margin
 - **Hair Toffee:** 77.78% margin
 - **Fizzy Lifting Drinks:** 60.00% margin
-

For these products, discontinuation may not be the best immediate action. Instead, management should assess **market demand, promotional potential, strategic relevance, and opportunities to increase sales**.

4. Retain / Review / Discontinue Classification

Based strictly on the profitability evidence, I recommend refining the file's two-category recommendation into three management categories:

Retain

-
- Wonka Bar - Scrumdiddlyumptious
 - Wonka Bar - Triple Dazzle Caramel
 - Wonka Bar - Milk Chocolate
 - Wonka Bar - Nutty Crunch Surprise
 - Wonka Bar - Fudge Mallows
-

Review

-
- Everlasting Gobstopper
 - Fizzy Lifting Drinks
 - Fun Dip
 - Hair Toffee
 - Laffy Taffy
 - Lickable Wallpaper
 - Nerds
 - Sweetarts
 - Wonka Gum
-

- **Kazookles — highest priority**
-

Discontinue

No product should be classified as definitively "Discontinue" based on this dataset alone.

The source file says "**Discontinuation Review**," not "**Discontinue**." This distinction is important for a professional business report. Actual discontinuation should also consider customer demand, strategic importance, inventory commitments, product lifecycle, and potential for pricing or cost improvements.

5. Key Business Insights

Insight 1 – Profitability is highly concentrated

Only **5 of 15 products (33.3%)** generate **95.06% of total gross profit**. This indicates substantial dependence on the core Wonka Bar portfolio.

Insight 2 – Core products combine scale and profitability

The retained Wonka Bar products generate **\$131,692.90 in sales** and **\$88,824.62 in gross profit**, demonstrating strong financial efficiency.

Insight 3 – Kazookles presents the strongest margin risk

Kazookles has only a **7.69% gross margin** and **92.31% cost-to-sales ratio**, making it the most obvious candidate for cost and pricing intervention.

Insight 4 – Low sales does not automatically mean poor profitability

Everlasting Gobstopper and Hair Toffee have margins of **80.00%** and **77.78%**, respectively. Their low sales volumes limit total contribution, but their underlying economics are attractive.

Insight 5 – Portfolio rationalization should be evidence-based

The analysis supports **reviewing** weaker products rather than automatically discontinuing them. Management should distinguish between **low-margin products** and **high-margin but low-demand products**.

6. Executive Summary

The Nassau Candy Product Line Profitability Analysis evaluates product-level sales, costs, gross profit, gross margin, unit volume, and cost-to-sales efficiency to identify the products that contribute most significantly to overall profitability.

The analysis demonstrates a highly concentrated profitability structure. **Five core Wonka Bar products generate \$88,824.62 in gross profit, representing 95.06% of total product profit**, while accounting for **92.88% of total sales**. Among these, **Wonka Bar - Scrumdiddlyumptious** generates the highest gross profit at **\$19,357.50**, while **Nutty Crunch Surprise** records the highest gross margin at **71.35%**.

At the other end of the portfolio, **Kazookles** represents the most significant financial risk, with only a **7.69% gross margin** and a **92.31% cost-to-sales ratio**. Other low-volume products also contribute relatively little to total profitability, although some demonstrate strong margins and should therefore be reviewed rather than automatically discontinued.

Overall, the findings support a **focused portfolio strategy**: protect and expand high-performing core products, investigate structurally weak-margin products, and evaluate low-volume products based on their future growth potential before making discontinuation decisions.

7. Conclusion and Management Recommendations

Conclusion

The profitability analysis demonstrates that Nassau Candy's product portfolio is **strong but highly concentrated**. A small number of core products generate the overwhelming majority

of revenue and gross profit, while several lower-volume products contribute minimally to overall profitability.

The analysis also highlights that **gross margin, sales volume, and total profit must be considered together**. A product with low sales may still have attractive unit economics, while a product with reasonable sales may have an unsustainably high cost structure.

Management Recommendations

1. Protect the core profitable products

Maintain strong inventory availability and pricing discipline for the five retained Wonka Bar products because they generate **95.06% of total profit**.

2. Prioritize Kazookles for intervention

Conduct a detailed **cost, pricing, and supplier analysis** for Kazookles because of its **7.69% margin and 92.31% cost-to-sales ratio**.

3. Review low-contribution products

Evaluate products such as Fun Dip, Nerds, Sweetarts, Laffy Taffy, and Wonka Gum based on demand, strategic relevance, and potential for improved profitability.

4. Do not immediately discontinue high-margin products

Products such as **Everlasting Gobstopper and Hair Toffee** should be assessed for sales-growth opportunities because their margins are strong despite low volumes.

5. Reduce portfolio dependency

Because five products generate **95.06% of total profit**, management should consider developing additional profitable products to reduce dependency on a small number of SKUs.

6. Establish ongoing profitability monitoring

A recurring product profitability dashboard should track **sales, gross profit, gross margin, cost-to-sales ratio, units sold, and profit contribution** to identify emerging margin risks early.

8. Validation of the Recommendations

Yes — the recommendations are broadly supported by the profitability data, with one important qualification.

The **Retain** recommendations are strongly supported because the five retained products collectively generate **\$88,824.62, or 95.06%, of total gross profit** and have gross margins between **64.92% and 71.35%**.

The **review** recommendation is particularly well supported for **Kazookles**, where the **7.69% margin and 92.31% cost-to-sales ratio** clearly indicate a structural profitability problem.

However, the label **"Discontinuation Review"** should not be interpreted as **"Discontinue."** Products such as Everlasting Gobstopper and Hair Toffee have excellent margins but very low sales. The profitability data alone cannot establish that discontinuation is the optimal decision.